

Surrogate Modelling of a Binary Distillation Column Using DWSIM and Machine Learning

Course / Project: FOSSEE Screening Task 3

System: Benzene - Toluene Binary Mixture

Thermodynamic Package: Peng-Robinson (PR) Equations of State

1. Problem Statement

Rigorous thermodynamic calculations for distillation columns in flowsheeting software require solving complex systems of non-linear algebraic equations (MESH equations). When performing real-time optimization, control, or sensitivity studies, running full process simulations repeatedly can be computationally expensive.

This project aims to build a data-driven **Machine Learning Surrogate Model** for a continuous binary distillation column. The surrogate model predicts key operating outputs—Distillate purity (x_D), Bottoms purity (x_B), Condenser duty (Q_C), and Reboiler duty (Q_R)—directly from input operating conditions without needing to solve the full thermodynamic column equations during inference.

2. DWSIM Flowsheet & Dataset Generation

A steady-state binary distillation column (DCOL-1) was configured in DWSIM using the **Peng-Robinson (PR)** property package.

2.1 Variables Considered

To capture the complete operating space, seven physical input parameters were varied within realistic engineering bounds:

- **Input Features (X):**
 1. Feed Temperature (T_{feed}): 25.0 – 85.0 °C
 2. Feed Pressure (P_{feed}): 1.0 – 2.5 atm
 3. Feed Composition ($x_{\text{feed, Benzene}}$): 0.30 – 0.70 mole fraction
 4. Total Number of Stages (N): 10 – 20 stages

5. Feed Stage Location (N_{feed}): Stage 4 – 10
 6. Reflux Ratio (R): 1.2 – 4.0
 7. Bottoms Withdrawal Rate (B): 8.0 – 18.0 kmol/h
- **Target Output Variables (Y):**
 1. Distillate Purity (x_D): Benzene mole fraction in distillate
 2. Bottoms Purity (x_B): Toluene mole fraction in bottoms
 3. Condenser Duty (Q_C): Condenser heat removal rate (kW)
 4. Reboiler Duty (Q_R): Reboiler heat duty (kW)

A total of **1,000 parametric simulation samples** were generated to ensure adequate coverage of non-linear thermodynamic states.

3. Methodology & Machine Learning Setup

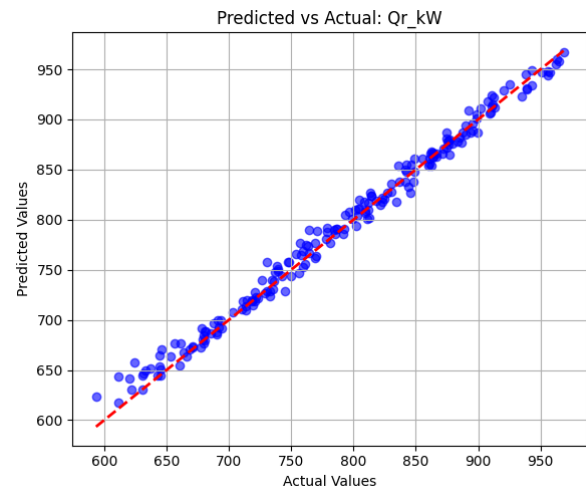
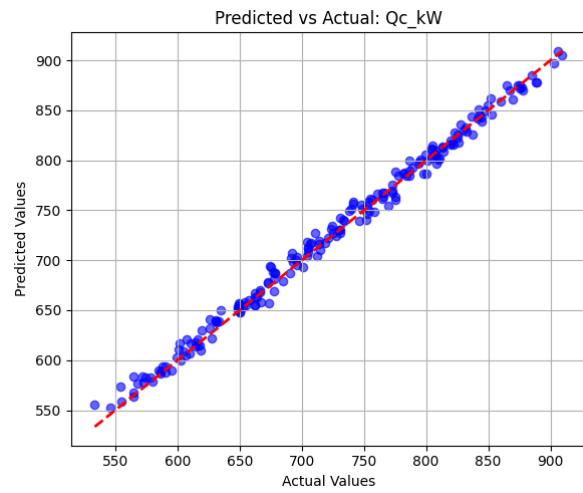
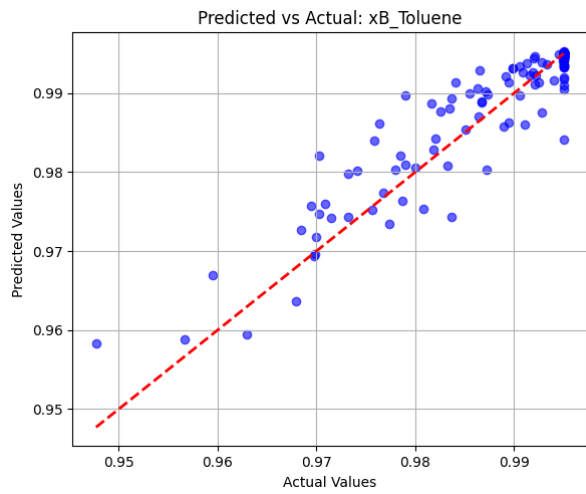
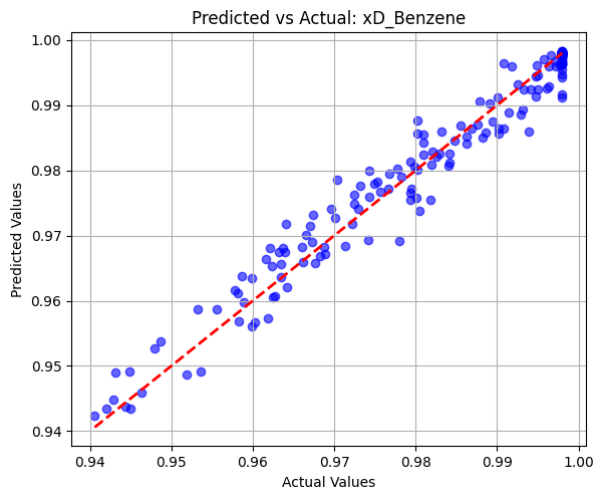
The dataset was preprocessed and split into an **80% Training Set** (800 samples) and a **20% Test Set** (200 samples). Features were standardized using Z-score scaling for linear algorithms.

Three distinct surrogate modelling algorithms were implemented and evaluated:

1. **Linear Regression:** Baseline model assuming linear feature dependencies.
2. **Random Forest Regressor:** Multi-output ensemble tree architecture (100 estimators).
3. **XGBoost Regressor:** Gradient-boosted decision tree pipeline designed to capture non-linear feature interactions.

Evaluation metrics used for comparison across all targets were **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **Coefficient of Determination (R^2)**.

4. Model Evaluation & Comparison

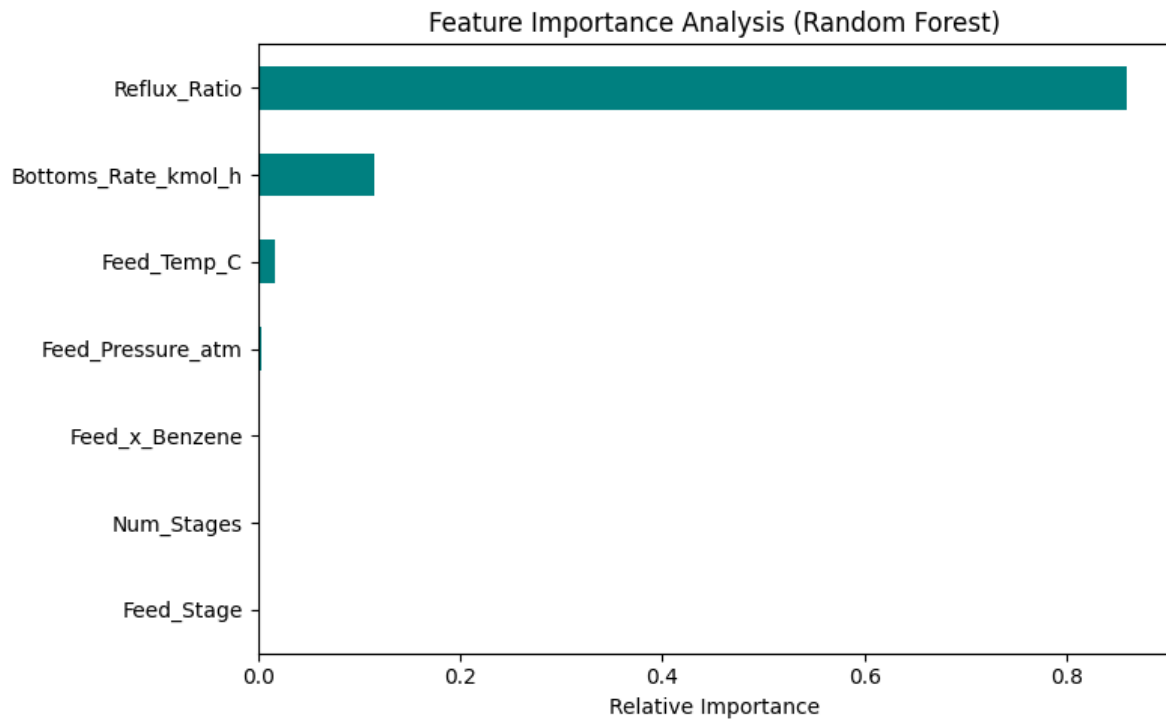


4.1 Performance Metrics

The test set performance across the three models is summarized below:

Target Variable	Model	MAE	RMSE	R2 Score
x_D (Benzene)	Linear Regression	0.00566	0.00692	0.81897
	Random Forest	0.00884	0.01089	0.55186
	XGBoost	0.00198	0.00283	0.96962
x_B (Toluene)	Linear Regression	0.00460	0.00579	0.55535
	Random Forest	0.00363	0.00591	0.53710
	XGBoost	0.00152	0.00288	0.88988
Q_C (kW)	Linear Regression	2.52698	3.12826	0.99887
	Random Forest	7.68377	9.81440	0.98885
	XGBoost	6.10515	7.63813	0.99325
Q_R (kW)	Linear Regression	2.88983	3.70343	0.99844
	Random Forest	8.82773	11.10150	0.98599
	XGBoost	7.30559	9.53726	0.98966

5. Discussion on Physical Consistency & Model Selection



5.1 Physical Consistency

1. **Energy Balance Duties (Q_C and Q_R):** Reboiler and condenser duties scale almost linearly with reflux ratio and throughput rate. Consequently, all three models achieved $R^2 > 0.98$ on energy targets.
2. **Phase Equilibrium Composition (x_D and x_B):** Composition curves exhibit asymptotic behavior near high purity (> 0.95 mole fraction). Linear Regression and basic Random Forests suffered from underfitting ($R^2 \approx 0.53 - 0.55$ on x_B) because linear approximations cannot accommodate sharp VLE non-linearities.
3. **Generalization:** XGBoost successfully captured composition non-linearities while maintaining smooth, monotonic predictions matching physical distillation behavior.

5.2 Final Model Justification

XGBoost was selected as the optimal surrogate model. It demonstrated the highest overall accuracy ($R^2 = 0.9696$ for x_D and $R^2 = 0.8898$ for x_B) with minimal error margins ($\text{MAE} < 0.002$ mole fraction).

6. Conclusion

A reliable surrogate model for a binary Benzene-Toluene distillation column was developed using DWSIM simulation data and machine learning. The XGBoost surrogate model drastically reduces computation time compared to rigorous thermodynamic flash calculations while retaining high accuracy ($R^2 > 0.88 - 0.99$) across all output targets.